

My Notes — Galaxy CodeVital Project Discussion

Quick Reference Map

Objectives

- O1 — Cold-start LV estimation for new students
- O2 — Session classification feeding the LV model
- O3 — Real-time LV inference during live sessions (<5s latency)
- O4 — Discrepancy detection (tutor report vs. student behavior)

Experiments

- 1.1 — Intake questionnaire as LV proxy → failed ($r=0.31$, too weak)
- 1.2 — First-session micro-signals via regression → failed (RMSE too high, $n=22$ too small)
- 1.3 — Bayesian prior initialization → most promising, converges in 2 sessions, needs $N \geq 50$
- 2.1 — Fine-tune Llama 3 8B on 200 transcripts → failed (61% vs 58% GPT-4, not significant)
- 2.2 — 90-second transcript segmentation → mixed (tutor-side improved, student-side degraded)
- 2.3 — RAG over labeled examples → best result (91% consistency, 66% accuracy, 2.1s latency)
- 3.1 — TVV feature ablation → 20 features → 8 features, <7% accuracy loss
- 3.2 — Streaming audio pipeline → 2.2s latency achieved, live accuracy 71% vs 84% recorded
- 3.3 — Adaptive per-session calibration → live accuracy 79%, target 85%, gap remains
- 4.1 — Ground truth dataset construction → 60 sessions, $\kappa=0.82$
- 4.2 — Behavioral signal classifier → precision 0.71, recall 0.54, insufficient
- 4.3 — Temporal pattern classifier → precision 0.78, recall 0.67, FP rate 22%, target <10%

8 TVV Features (from 3.1) Speech pace, question cadence, explanation repetition, student silence duration, tutor-to-student speaking ratio, vocabulary complexity (TTR), correction frequency, session energy

Open into FY2026

- Cold-start needs $N \geq 50$ validation
- Live accuracy gap: 79% achieved, 85% needed
- Discrepancy FP rate: 22% current, <10% needed
- RAG at scale: unknown behavior at 2000+ examples

- AI Tutor Avatar: zero implementation, fully open
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⚠ Do Not Propose as New Ideas

Bayesian cold-start, RAG over labeled examples, 8-feature TVV, post-session batch, real-time pipeline — already built.

Opening

The FY2025 work is substantial. The ablation study on the TVV, the RAG-over-examples finding in 2.3, the segmentation asymmetry between tutor-side and student-side signals — these are genuinely novel findings. I want to talk about where I think I can contribute on the open problems.

Cold-Start (O1 / 1.3)

The Bayesian approach is the right call — 1.1 correctly eliminated survey-based initialization and the convergence in 2 sessions is a strong result. The blocker is statistical: 22 students isn't enough to deploy with confidence.

My question is whether the prior distributions are stratified finely enough. Grade level is a proxy — but developmental differences in self-regulation between a 9-year-old and a 16-year-old are significant. Age-stratified priors might generalize better than grade-stratified ones, especially as the validation cohort grows toward $N \geq 50$.

Session Classification (O2 / 2.3)

The RAG-over-examples finding is the most important architectural decision in the doc. Consistency at 91% matters more than raw accuracy here — an inconsistent classifier introduces noise that compounds into every LV score downstream. Eliminating fine-tuning at this data scale was the right call.

The segmentation asymmetry from 2.2 — tutor-side signals can be segmented, student-side needs full session context — should probably drive two separate retrieval pipelines, not one. As the labeled dataset scales toward 2000+ examples, the architecture may need to split rather than treating all four classification tasks the same way.

Real-Time Accuracy Gap (O3 / 3.3)

79% live vs. 84% recorded — that's a noise normalization problem. The static 2-minute calibration from 3.3 can't capture how noise characteristics drift throughout a session. Two directions I'd explore:

One — rolling window normalization that updates feature statistics continuously rather than anchoring to the first 2 minutes.

Two — replacing the noise-sensitive acoustic features with transcript-derived equivalents. Session energy and volume variance from the 8-feature set collapse with bad microphones. Whisper-diarized transcripts give you turn-taking interruption count — a stable proxy that doesn't degrade with audio quality.

Also worth asking: is 85% a hard product requirement for real-time, or is there room for a confidence-flagged hybrid? Concretely — instead of forcing a point estimate at 79% accuracy regardless of conditions, the pipeline outputs a score plus a confidence interval derived from how stable the 8-feature readings are within the rolling window. Noisy microphone, erratic features → wide interval → suppress real-time, defer to batch. Clean session → narrow interval → show live feedback. The system self-regulates. The 85% target becomes a threshold for when to surface real-time output, not a blanket requirement across all sessions.

Discrepancy Detection (O4 / 4.3)

The temporal signature finding in 4.3 is the key insight — progressive engagement degradation is a different signal from uniformly low engagement. The gap to <10% false positive rate is a confounding problem. The classifier can't separate tutor-caused disengagement from structural disengagement caused by session fatigue, topic difficulty, or time of day.

I'd add session-context features to condition on those confounders — session number in the week, subject, time slot, student's historical engagement baseline. The 60-session dataset is workable given the $\kappa=0.82$ agreement — the issue is feature engineering, not dataset size.

AI Tutor Avatar (FY2026 Open)

This is the one area with zero implementation yet and where I have the most concrete thoughts.

The core problem they've named correctly — generic LLMs don't represent tutor personality. The Avatar needs to respond the way Tutor Sarah actually teaches, not the way a model trained on the internet teaches.

RAG over that tutor's session transcripts is the right first architecture. Student asks about fractions — retrieve Sarah's past explanations of fractions — generate in her style. This doesn't

require training a new model. It grounds responses in her actual language. Works with existing session data. Prototype in weeks, not months.

Two honest limitations worth naming: per-tutor RAG only works once a tutor has enough session history — new tutors have the same cold-start problem as students. And voice synthesis is a separate, harder problem — text Avatar can ship before voice is solved.

On privacy: at Future Focused Education I deployed local model servers specifically because parents were worried about student data reaching large AI companies — exact same concern, exact same architecture. At HyperAnalytics I built the PII stripping pipeline for HIPAA compliance. COPPA and PIPEDA require the same pattern — on-premise inference, PII stripped before any model touches the data.

Questions to Ask

- "For cold-start — where does the $N \geq 50$ validation cohort stand, is student recruitment ongoing?"
 - "On real-time — is 85% a hard product requirement or is there room for a confidence-flagged hybrid?" (*see real-time section above for what that looks like*)
 - "For the Avatar — is the priority text responses in the tutor's style, or does voice synthesis need to be part of v1?"
 - "Why real-time at all — who sees the estimate mid-session and what specific action do they take on it? If the answer is a post-session debrief or parent report, batch at 84% is strictly better and already solved."
 - "The matching engine was last addressed in FY2024 — cosine similarity and RL both failed. Is that work blocked on LV inputs being solved first, or is it a parallel open problem?"
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Additional Observations

Matching engine is absent from FY2025 The entire FY2025 doc is about LV inputs. But the matching engine — which all of this feeds — was last mentioned in FY2024 where cosine similarity and RL both failed. No matching experiments in FY2025. Either it's blocked on LV inputs or it's a separate open problem nobody is actively working on. Worth understanding before proposing anything there.

Bayesian cold-start has a circular validation problem The prior distributions in 1.3 were derived from the same 22-student dataset used to validate the approach. That's not a true validation — you're testing a model whose priors were built from the same data. A real $N \geq 50$ validation needs a held-out cohort that had no influence on prior construction. Worth flagging gently.

